BICOL UNIVERSITY

College of Engineering | Department of Chemical Engineering

ASYNCHRONOUS ACTIVITY NO. 1

GitHub Portfolio Setup and Three Organic Compound Infographics

CHEM 13 - Organic Chemistry (Lecture) | BS Chemical Engineering

Activity mode	Individual, asynchronous	**Main output**	3 infographics + GitHub portfolio
Submission	Public GitHub repository link	**Primary folder**	001_structure_bonding_lastname

• Purpose of the Activity

This first asynchronous activity establishes your individual CHEM 13 GitHub portfolio and begins the semester with a short scientific-communication exercise. You will create the required repository and folder structure, then prepare three separate infographics - one infographic for each of three organic compounds that you personally find interesting, useful, unusual, or important. The activity introduces the habits expected throughout the course: correct chemistry, organized digital work, evidence-based claims, proper referencing, and clear communication.

• Learning Objectives

- Create a professional GitHub account and a public CHEM 13 portfolio repository.
- Organize the repository using the required course folder structure and naming convention.
- Identify three organic compounds and accurately communicate their molecular identity, structure, properties, useful or remarkable features, risks, and relevance.
- Use credible scientific sources and provide APA-style references for factual claims and data.
- Demonstrate concise visual communication appropriate for undergraduate chemical engineering.

• What You Will Submit

Your submission has two connected parts:

- **Part A - GitHub Portfolio Setup:** create your account, public repository, and required folder structure.
- **Part B - Three Organic Compound Infographics:** select three different organic compounds and prepare one complete infographic for each compound.

• Part A - GitHub Account and Portfolio Setup

Step 1. Create a GitHub account

- Go to GitHub (https://github.com/) and create a free account if you do not already have one.
- Use a professional and recognizable username whenever possible.
- Verify your email address and sign

in to your account. **Step 2. Create a**

public CHEM 13 repository

Repository name: `chem13-portfolio-lastname`

Example: chem13-portfolio-santos

Step 3. Create the required folder structure

All three infographics for Activity No. 1 must be saved inside:
`001_structure_bonding_lastname/001_infographics_exercise_lastname`

• Part B - Three Organic Compound Infographics

Select THREE DIFFERENT organic compounds. Prepare ONE infographic for EACH compound. Therefore, your final output for this part is three separate infographics.

Choosing your three compounds

Choose compounds that genuinely interest you. They may come from food, medicines, fuels, fragrances, polymers, cosmetics, agriculture, natural products, consumer products, or industrial processes. You are encouraged to choose compounds with clearly different functions or applications so that your three outputs show variety.

- Possible examples: caffeine, ethanol, acetic acid, acetone, aspirin, vanillin, limonene, glycerol, ethylene, lactic acid, citric acid, benzene, urea, or another instructor-approved organic compound.

Required content of EACH infographic

No.	Content Area	What to Show
1	**Compound identity**	Common name, IUPAC name, molecular formula, and molar mass.
2	**Structure**	A correct structural formula or skeletal structure. Identify the major functional group(s) and, when useful, a notable structural feature.
3	**Important properties**	Select the most meaningful physical and/or chemical properties - for example state, appearance, polarity, solubility, melting/boiling behavior, acidity/basicity, reactivity, or intermolecular-force behavior. Do not crowd the infographic with unnecessary data.
4	**What is amazing about it?**	Explain one memorable, surprising, or scientifically interesting thing the compound can do or why it matters. This should be more than a slogan: briefly explain the chemistry behind it.
5	**Uses and value**	Show important everyday, medical, commercial, industrial, or societal uses. Where relevant, mention why the compound is valuable to a process or product.

No.	Content Area	What to Show
6	**Risks and safety**	Identify significant health, fire, reactivity, handling, environmental, or disposal risks. Include only risks that actually apply to the chosen compound.
7	**Chemical engineering connection**	Add at least one concise connection to chemical engineering, such as reaction engineering, separation, extraction, crystallization, distillation, formulation, process safety, waste treatment, or product design.
8	**References**	Provide the sources for factual claims and data. Use APA-style references in a small but readable footer and provide the full references again in your repository README or reference file.

Infographic design and file requirements

- Create three separate infographic files - not one poster divided into three compounds.
- Recommended size: one-page portrait or landscape format. Canva, PowerPoint, Google Slides, Adobe Express, or similar tools may be used.
- Keep text concise. Use headings, short statements, icons/diagrams, and visual hierarchy instead of long paragraphs.

- The molecular structure must be scientifically correct. You may draw it using Avogadro, BKChem, Ketcher, or another appropriate chemistry tool.
- APA references may be placed in smaller type at the bottom of the infographic; approximately 7-9 pt is acceptable provided the text remains readable in the exported file.
- Export each infographic as PNG, JPG, or PDF. Also retain your editable source file when practical.

Required file naming

• Worked Example - What One Infographic Could Contain

The example below uses CAFFEINE only to demonstrate the expected depth and organization. Do not copy the wording as your own work; use it as a guide for how much information to include and how to connect chemistry to significance. See attached appendix for an actual example.

Headline	CAFFEINE - The Molecule Behind the Morning Boost
Identity	Common name: Caffeine | IUPAC name: 1,3,7-trimethylpurine-2,6-dione | Formula: C8H10N4O2 | Molar mass: about 194.19 g/mol.
Structure	Show the skeletal structure of caffeine. Point out the two carbonyl groups and the nitrogen-containing fused-ring system; describe caffeine as a methylxanthine compound.
Key properties	White crystalline organic compound; relatively polar; soluble in water, with solubility increasing at higher temperature. Its molecular structure allows several hydrogen-bond acceptor sites but no O-H or N-H hydrogen-bond donor group.

Amazing feature	Caffeine can make us feel more alert because it interferes with adenosine signaling in the nervous system. Instead of merely writing "caffeine wakes you up," connect the effect to the molecule's interaction with biological receptors.
Uses/value	Naturally present in coffee, tea, and cacao; also used in beverages and some pharmaceutical preparations. It has commercial value in food, beverage, and pharmaceutical products.
Risks	Excessive intake may produce unwanted effects such as restlessness, sleep disruption, increased heart rate, or other adverse responses. The infographic should avoid sensational claims and should use an authoritative safety source.
Chemical engineering link	Caffeine can be connected to extraction and separation operations: chemical engineers may encounter solvent extraction, solid-liquid extraction, adsorption, crystallization, purification, process control, and product formulation when working with caffeine-containing streams or products.
Suggested "wow" line	Small molecule, big process story: caffeine connects molecular structure to extraction, separations, formulation, consumer products, physiology, and safety.
Reference footer example	National Center for Biotechnology Information. (n.d.). PubChem compound summary for CID 2519, Caffeine. PubChem. EFSA Panel on Dietetic Products, Nutrition and Allergies (NDA). (2015). Scientific opinion on the safety of caffeine. EFSA Journal, 13(5), 4102. https://doi.org/10.2903/j.efsa.2015.4102

Suggested one-page layout for each infographic

At the very bottom: add a compact APA reference footer (and an AI-use disclosure note if applicable).

• Sources and APA Referencing

Use credible, traceable scientific sources. Suitable sources include textbooks, peer-reviewed journal articles, PubChem, NIST, government or university resources, recognized safety data sheets, and reputable professional/scientific organizations.

- Every numerical property, safety statement, unusual fact, or specialized claim should be traceable to a source.
- Use short APA-style references at the bottom of the infographic. When space is limited, keep them compact but readable.
- Create a references_lastname.md file in the activity folder containing the complete APA 7th edition references used across all three infographics.
- Do not use random blogs, social-media posts, or unattributed online graphics as scientific evidence.

Use of Artificial Intelligence

AI tools may be used as learning and communication support, consistent with the CHEM 13 syllabus, but the student remains responsible for the correctness of every submitted claim, structure, reference, and interpretation.

- Permitted examples: brainstorming possible compounds, suggesting an infographic outline, improving wording, or helping identify questions to research.
- Required: independently verify chemistry, numerical data, risks, and references using credible sources.
- Disclosure: create ai_use_disclosure_lastname.txt and state the AI tool used and what it was used for. If no AI was used, state that explicitly.

Final Folder Contents and Submission

Before submitting your repository link through Google Classroom, the activity folder should look similar to the example below:

- Confirm that the repository is PUBLIC and accessible without requesting permission.
- Open each infographic from GitHub to confirm that the file displays correctly.
- Submit only the GitHub repository link in the designated Google Classroom activity unless additional instructions are posted.

Assessment Rubric

Criterion	Description	Weight
GitHub setup & organization	Public repository created correctly; required folders and filenames are present and organized.	**15%**
Chemistry accuracy	Structures, names, properties, functional groups, uses, risks, and explanations are scientifically correct.	**30%**
Content quality & depth	Each of the three infographics meaningfully covers identity/structure, important properties, an "amazing" feature, uses/value, risks, and a chemical engineering connection.	**25%**
Visual communication	Infographics are readable, concise, logically arranged, visually engaging, and appropriate for scientific communication.	**15%**
Sources & APA referencing	Claims are supported by credible sources; reference footer is present and the repository contains complete APA references.	**10%**
Compliance & AI disclosure	Three separate files are submitted in the correct folder and AI use is disclosed as required.	**5%**
TOTAL		**100%**

Pre-Submission Checklist

- I created or verified my GitHub account.

- I created a public repository named chem13-portfolio-lastname.

- I created the complete CHEM 13 folder structure.
- I created 001_structure_bonding_lastname/001_infographics_exercise_lastname.
- I selected three DIFFERENT organic compounds.
- I created one complete infographic for each compound (three infographics total).
- Each infographic includes structure/identity, relevant properties, an amazing or significant feature, uses/value, risks/safety, and a chemical engineering connection.
- I verified my chemical structures, data, and factual claims using credible sources.
- I included readable APA-style references in each infographic and complete references in references_lastname.md.
- I included ai_use_disclosure_lastname.txt.
- I opened all files from GitHub and checked that they display properly.
- I submitted the public GitHub repository link through Google Classroom.

The goal is not to fit everything you can find about a compound onto one page. The goal is to select accurate, meaningful information and communicate why the molecule is chemically interesting and relevant.

APPENDIX: Example Infographic

EXAMPLE INFOGRAPHIC

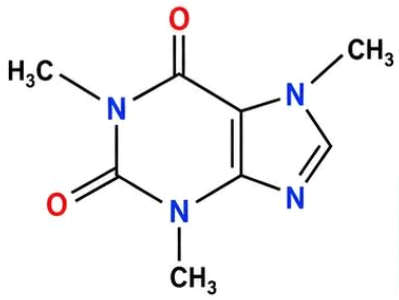
Caffeine

An example for CHEM 13 students

IUPAC name:
1,3,7-trimethylxanthine

Molecular formula:
$C_8H_{10}N_4O_2$

Molar mass:
194.19 g/mol



Functional groups:
amide-like carbonyls, imide/nitrogen-containing heterocycle

One of the world's most studied and consumed natural compounds!

1. Structure & Identity

- **Common name:** Caffeine
- **Family:** Alkaloid
- **Appearance:** White crystalline solid
- **Natural sources:** Coffee beans, tea leaves, cacao

2. Important Properties

- Slightly bitter organic compound
- Water-soluble; more soluble in hot water
- Melting point: about 235–238 °C
- Contains multiple nitrogen atoms that affect polarity and intermolecular interactions
- Biologically active stimulant

3. What Makes It Amazing?

- Blocks adenosine receptors, helping reduce drowsiness
- Widely used in beverages, medicines, and food products
- A small molecule with a big effect on alertness and human behavior

4. Uses / Value

- Used in coffee, tea, soft drinks, and energy drinks
- Added to some headache and cold medications
- Valuable commercial compound in the food and pharmaceutical industries
- Example of a naturally occurring organic compound with major market value

5. Risks & Safety

- Too much caffeine may cause anxiety, rapid heartbeat, insomnia, or restlessness
- Moderate intake is generally tolerated, but sensitivity varies among people
- Concentrated caffeine products should be handled carefully
- Excess consumption may pose health risks

6. Chemical Engineering Relevance

- Relevant to extraction and separation processes
- Important in food processing and quality control
- Useful example for discussing solubility, purification, and product formulation
- Connects organic chemistry to industrial applications

7. Design Notes for Students

- Use one compound per infographic
- Include structure, properties, uses/value, amazing feature, and risks
- Cite reliable sources in small font
- Keep the layout neat, accurate, and visually balanced

Example APA References

- National Center for Biotechnology Information. (2024). *PubChem compound summary for caffeine.*
- Smith, J. (2020). *Organic chemistry fundamentals for students.* Academic Press.
- U.S. Food and Drug Administration. (2023). *Spilling the beans: How much caffeine is too much?*

Chemical Spotlight: CAFFEINE

The Molecule That Powers the World

THE IDENTITY

Common Name:
Caffeine

IUPAC Name:
1,3,7-Trimethylpurine-2,6-dione

Chemical Formula:
$C_8H_{10}N_4O_2$

Classification:
Alkaloid (Natural Nitrogen-Containing Compound)

KEY PROPERTIES

Molar Mass:
194.19 g/mol

Appearance:
White, odorless crystalline powder (pure state).

Taste:
Distinctly bitter taste profile.

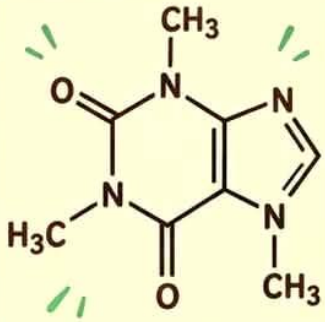
Solubility:
Moderately soluble in water; increases significantly with heat.

CHEMICAL ENGINEERING LINK



Industrial Extraction:
Supercritical Fluid Extraction (SFE) using carbon dioxide (CO_2) at critical pressure/ temperature to selectively extract caffeine from green coffee beans without leaving toxic solvent residues (Decaffeination process).

STRUCTURE



Key Feature:
Structurally similar to adenosine (allows receptor-binding competitive inhibition).

USES / VALUE & WHERE TO FIND IT

Primary Uses:
- Central nervous system stimulant
- Alertness enhancement
- Fatigue reduction

Natural Sources:


Coffee beans


Tea leaves


Cacao pods

Synthesized Sources:


Energy drinks


Sodas


Pharmaceuticals

How It Works (The Brain Hijack):
Adenosine builds up in the brain to signal fatigue. Caffeine's molecular shape mimics adenosine, docking into adenosine receptors to block sleepiness signals and keep you alert.

WOW! TRIVIA

Nature's Defense:
Plants don't make caffeine for human energy—they synthesized it as a natural neurotoxic pesticide to paralyze insects that try to eat them!

Safe Daily Limit:
400 mg/day
(approx. 4 brewed coffee cups for healthy adults).

CHEMICAL SPOTLIGHT:

ASPIRIN

The Modern Miracle from Ancient Bark

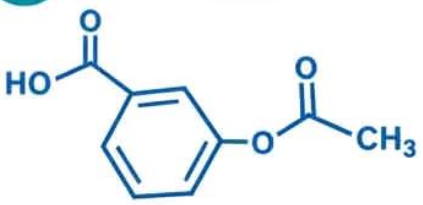
THE BASICS

Common Name	:	Aspirin
IUPAC Name	:	2-Acetyloxybenzoic acid
Chemical Formula	:	$C_9H_8O_4$
Classification	:	Nonsteroidal Anti-inflammatory Drug (NSAID) / Ester / Salicylate

KEY PROPERTIES

Molar Mass	:	180.16 g/mol
Melting Point	:	135 °C (275 °F)
Appearance	:	White, crystalline powder or compressed tablets.
Solubility	:	Poorly soluble in water; dissolves readily in ethanol and alkaline solutions.

STRUCTURE



Key Feature:
Acetyl group transfer mechanism responsible for irreversible enzyme inhibition.

WHERE TO FIND IT



Over-the-counter painkiller tablets
(modern form)



Willow tree bark
(natural source of salicin)

USES & HOW IT WORKS

Primary Uses

- Analgesic (pain relief)
- Antipyretic (fever reduction)
- Anti-inflammatory
- Anticoagulant (blood thinner)

Modern Form

Over-the-counter painkiller tablets and low-dose cardiovascular therapies.

Ancient Origin

Willow tree bark (natural source of salicin).

How It Works

(Stopping Pain at the Source)

When tissue is damaged, the body produces prostaglandins that cause swelling and pain signals. Aspirin irreversibly acetylates and inhibits COX enzymes (COX-1 and COX-2), shutting down prostaglandin synthesis at the source.



Tissue damage → prostaglandins (produce pain & swelling)

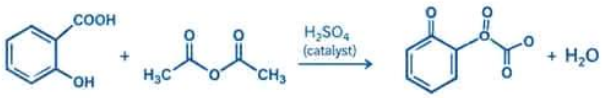
Aspirin inhibits COX-1 & COX-2 (irreversible)

Less prostaglandin = less pain & inflammation

CHEMICAL ENGINEERING LINK

Industrial Synthesis

Acetylation reaction of salicylic acid with acetic anhydride using a sulfuric acid (H_2SO_4) catalyst. The process involves precise temperature control, crystallization, filtering, and drying to ensure pharmaceutical purity.



Salicylic acid + **Acetic anhydride** → **Aspirin**

QUICK FACTS & TRIVIA

Ancient Roots

"Long before pills existed, Hippocrates and ancient Egyptian physicians prescribed brewed willow bark tea to ease labor pains and fevers!"

Heart Saver

"A low daily dose (81 mg) prevents blood platelets from clumping together, drastically reducing the risk of secondary heart attacks and ischemic strokes."

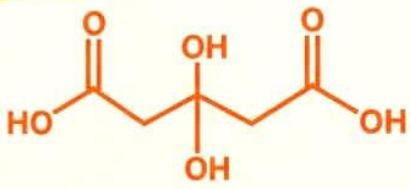
Chemical Spotlight: CITRIC ACID

Nature's Sour Powerhouse & Cellular Engine

The Basics

Common Name:	Citric Acid
IUPAC Name:	2-hydroxypropane-1,2,3-tricarboxylic acid
Chemical Formula:	$C_6H_8O_7$
Classification:	Weak Organic Carboxylic Acid / Tricarboxylic Acid

Chemical Structure



$C_6H_8O_7$

Tricarboxylic structure responsible for strong chelating ability and pH modification.

Key Properties

- **Molar Mass:** 192.12 g/mol
- **Appearance:** White, crystalline powder or translucent granules.
- **Taste Profile:** Intensely tart and sharp sour taste.
- **Solubility:** Extremely soluble in water; highly hydrophilic.

Where to Find It & Uses

Natural Sources



- Lemons and limes (highest concentrations)
- Oranges and grapefruits

Commercial & Everyday Uses



- Sour gummy candies (outer acid dusting)
- Skincare products (pH level balancer & AHA exfoliant)

Flavor Enhancer



Triggers sour taste receptors on the tongue, adding a crisp, refreshing bite.

Nature's Preservative



Lowers overall food pH to inhibit microbial, bacterial and mold growth, extending shelf life naturally.

Chemical Engineering Link



Industrial Fermentation

Mass production uses submerged fermentation of molasses or corn steep liquor using the **fungus *Aspergillus niger*, followed by calcium hydroxide** precipitation and sulfuric acid recovery.



WOW FACTS & TRIVIA

Biological Engine

Citric Acid is the central intermediate in the **Krebs Cycle** (Citric Acid Cycle)—the fundamental metabolic process living cells use to produce ATP cellular energy!

Household Magic

It acts as a natural chelating agent, making it **exceptionally good at** dissolving limescale, rust, and hard water minerals from kettles and faucets without harsh chemical fumes.